VTEAM Memristor Model Calculations

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1 Equations and Parameters

1.1 Equations

1. $R = R_{ON}w + R_{OFF}(1-w)$
2. $f(w) = 1 - (2w-1)^{20}$ (Biolek window function)
3. $$f(V) = \begin{cases} K_{ON}(\frac{V}{V_{ON}})^{\alpha_{ON}}, & V < V_{ON} \\ 0, & V_{OFF} \leq V \leq V_{ON} \\ K_{OFF}(\frac{V}{V_{OFF}})^{\alpha_{OFF}}, & V > V_{OFF} \end{cases} \quad (1)$$
4. $w_{n+1} = w_n + \frac{dw}{dt}$
5. $\frac{dw}{dt} = f(V)f(w)$

1.2 Parameters

1. $V_{ON} = -0.15V$, $V_{OFF} = +0.2V$
2. $K_{ON} = -0.02$, $K_{OFF} = +0.02$
3. $\alpha_{ON} = \alpha_{OFF} = 2$
4. $R_{ON} = 100\Omega$, $R_{OFF} = 16k\Omega$
5. $w_{init} = 6554(0.1)$ $R = 16000 + 15900w$

2 Calculations

For waveforms, refer to Satyadutt's hand draw timing diagram. This document will contain only calculations and validation of the VTEAM model.

1. **0 - 5 ns.**
   Reset (rst) is enabled. The next clock edge where reset is released is 15 ns., $\therefore V = 0V$.

2. **15 - 30 ns.**
   $w = w_{init} = 6554 = 0.1$, $R = 14410\ \Omega$

3. **30 - 50 ns.**
   V = 13107 = 0.2 V = V_{OFF} $f(V) = K_{OFF}(1-1)^{\alpha_{OFF}} = 0$ $\frac{dw}{dt} = 0$; No change in state variable $\therefore w = 0.1, R = 14410\Omega$

   We see whenever $V = V_{OFF}$ or V_{ON}, the state is the same and memristance is the same. $\therefore V > V_{OFF} / < V_{ON}$. 50 - 55 ns. : Negedge clk, no change in state

4. **55 - 60 ns.**
   V = 19661 = 0.3 V, $w_n = 0.1$; $V > V_{OFF}$
   $f(V) = 0.02[\frac{0.3}{0.2} - 1]^2 = 0.005$
   $f(w) = 1 - (2w-1)^{20} = 0.9884$
   $\frac{dw}{dt}$ =0.005 x 0.9884 = 0.00492
   w_{n+1}= 0.10492 = 6877
   R = 16k - 15.9k(0.10492) = 14331 Ω

   60 - 65 ns. : Negedge clock

5. **65 - 70 ns.**
   V = 19661 = 0.3 V, $w_n = 0.10492$; $V > V_{OFF}$
   $f(V) = 0.02[\frac{0.3}{0.2} - 1]^2 = 0.005$
   $f(w) = 1 - (2w-1)^{20} = 0.9909$
   $\frac{dw}{dt}$ =0.005 x 0.9909 = 0.004954
   $w_{n+1} = 0.10987 = 7201$
   R = 16k - 15.9k(0.10987) = 14253 Ω

6. **70 - 90 ns.**
   V = 0 V, $\frac{dw}{dt} = 0$
   $\therefore$ No change in state, previous w, R values are retained.

7. **90 - 115 ns.**
   V = -0.15 V = V_{ON}, $\frac{dw}{dt} = 0$ (State retained)
   110 - 115 ns. : Negedge clock

8. **115 - 125 ns.**
   V = -13107 = -0.2 V, $w_n = 0.10987$; $V < V_{ON}$
   $f(V) = 0.02[\frac{0.3}{0.2} - 1]^2$ = -0.00222
   $f(w) = 1 - (2w - 1)^{20} = 0.993$
   $\frac{dw}{dt}$ =0.005 x 0.993 = -0.002206
   $w_{n+1} = 0.10766 = 7056$
   R = 16k - 15.9k(0.10766) = 14288 Ω

9. **125 - 130 ns.**
   V = -13107 = -0.2 V, $w_n = 0.10987$; $V < V_{ON}$
   $f(V) = 0.02[\frac{0.3}{0.2} - 1]^2$ = -0.00222
   $f(w) = 1 - (2w - 1)^{20} = 0.99216$
   $\frac{dw}{dt}$ =0.005 x 0.99216 = -0.0022045
   $w_{n+1} = 0.10545 = 6911$
   R = 16k - 15.9k(0.10766) = 14323 Ω

10. **130 - 155 ns.**
    V = 0 V, $\frac{dw}{dt} = 0$
    $\therefore$ No change in state, previous w, R values are retained.
    150 - 155 ns. : Negedge clock

11. **155 -165 ns.**
    V = 19661 = 0.3 V, $w_n = 0.10545$; $V > V_{OFF}$
    $f(V) = 0.02[\frac{0.3}{0.2} - 1]^2 = 0.005$
    $f(w) = 1 - (2w - 1)^{20} = 0.99123$
    $\frac{dw}{dt}$ =0.005 x 0.99123 = 0.0049561
    $w_{n+1} = 0.1104 = 7235$
    R = 16k - 15.9k(0.1104) = 14244 Ω

12. **165 - 175 ns.**
    V = 19661 = 0.3 V, $w_n = 0.1104$; $V > V_{OFF}$
    $f(V) = 0.02[\frac{0.3}{0.2} - 1]^2 = 0.005$
    $f(w) = 1 - (2w - 1)^{20} = 0.99319$
    $\frac{dw}{dt}$ =0.005 x 0.99319 = 0.0049659
    $w_{n+1} = 0.11536 = 7560$
    R = 16k - 15.9k(0.11536) = 14165 Ω

13. **175 - 185 ns.**
    V = 19661 = 0.3 V, $w_n = 0.11536$; $V > V_{OFF}$
    $f(V) = 0.02[\frac{0.3}{0.2} - 1]^2 = 0.005$
    $f(w) = 1 - (2w - 1)^{20} = 0.99473$
    $\frac{dw}{dt}$ =0.005 x 0.99473 = 0.0049736
    $w_{n+1} = 0.12033 = 7885$
    R = 16k - 15.9k(0.12033) = 14086 Ω

14. **185 - 195 ns.**
    V = 19661 = 0.3 V, $w_n = 0.12033$; $V > V_{OFF}$
    $f(V) = 0.02[\frac{0.3}{0.2} - 1]^2 = 0.005$
    $f(w) = 1 - (2w - 1)^{20} = 0.99593$
    $\frac{dw}{dt}$ =0.005 x 0.99593 = 0.0049561
    $w_{n+1} = 0.12033 = 7235$
    R = 16k - 15.9k(0.12033) = 14007 Ω

15. **195 - 205 ns.**
    V = -13107 = -0.2 V, $w_n = 0.12033$; $V < V_{ON}$
    $f(V) = 0.02[\frac{0.3}{0.2} - 1]^2$ = -0.00222
    $f(w) = 1 - (2w - 1)^{20} = 0.99688$
    $\frac{dw}{dt}$ =0.005 x 0.99216 = -0.002215
    $w_{n+1} = 0.123094 = 8067$
    R = 16k - 15.9k(0.123094) ≈ 14043 Ω

16. **205 - 215 ns.**
    V = -13107 = -0.2 V, $w_n = 0.123094$; $V < V_{ON}$
    $f(V) = 0.02[\frac{0.3}{0.2} - 1]^2$ = -0.00222
    $f(w) = 1 - (2w - 1)^{20} = 0.99649$
    $\frac{dw}{dt}$ =0.005 x 0.99216 = -0.002214
    $w_{n+1} = 0.12088 = 7922$
    R = 16k - 15.9k(0.12088) ≈ 14078 Ω

17. **215 - 225 ns.**
    V = -13107 = -0.2 V, $w_n = 0.12088$; $V < V_{ON}$
    $f(V) = 0.02[\frac{0.3}{0.2} - 1]^2$ = -0.00222
    $f(w) = 1 - (2w - 1)^{20} = 0.99605$
    $\frac{dw}{dt}$ =0.005 x 0.99216 = -0.0022123
    $w_{n+1} = 0.11866 = 7776$
    R = 16k - 15.9k(0.11866) ≈ 14113 Ω

18. **225 - 250 ns.**

    V = 0 V, state is retained
    $\therefore w = 0.11866, R = 14113\Omega$